

COMPUTER ORIENTED STATISTICAL METHODS

II B. TECH- I SEMESTER

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6BS03	BSC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	40	60	100

COURSE OBJECTIVES

1. Evaluation of Probability distribution of Discrete and Continuous random variables and their moments.
2. The concept of correlation and regression, covariance and sampling distribution
3. Evaluation of the given data for appropriate test of hypothesis for large samples.
4. Evaluation of the given data for appropriate test of hypothesis for small samples and one way ANOVA
5. To learn the concept of Markov chain, transition probabilities in discrete & continuous time and Stochastic simulation techniques.

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Evaluation of Probability distribution of Discrete and Continuous random variables and their moments.
2. Apply the concept of correlation and regression to find covariance and Sampling distribution
3. Evaluate the given data for appropriate test of hypothesis for Large samples.
4. Evaluate the given data for appropriate test of hypothesis for small samples and one way ANOVA
5. Recognize if a given stochastic system with finite number of states is a Markov chain or not and also identify classes of states in Markov chains and Characterize the states.

UNIT - I	RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS	CLASSES: 08
Random Variables – Discrete and Continuous. Probability distributions, mass function/ density function of a probability distribution, mathematical expectation,. Binomial, Poisson, Normal distributions -their Properties. Moments about origin, central moments, skewness, Kurtosis and find the mean and variance.		
UNIT - II	CORRELATION & REGRESSION AND SAMPLING DISTRIBUTIONS	CLASSES: 09
Coefficient of correlation, the rank correlation, Covariance of two random variables. Regression- Regression Coefficient, The lines of regression. Sampling: Definitions of population, sampling, statistic, parameter. Types of sampling, Expected values of Sample mean and variance, sampling distribution, Standard error, Sampling distribution of means and sampling distribution of variance. Parameter estimation- Point estimation and interval estimation		
UNIT - III	TESTING OF HYPOTHESIS - I	CLASSES: 09
Testing of hypothesis: Null hypothesis, Alternate hypothesis, Type I & Type II errors – critical region, confidence interval, Level of significance. One sided test, Two sided test. Large sample tests: (i) Test of Equality of means of two samples, equality of sample mean and population mean (cases of known variance & unknown variance, equal and unequal variances) (ii) Tests of significance of difference between sample S.D and population S.D. (iii) Tests of significance difference between sample proportion and population proportion, difference between two sample proportions.		
UNIT – IV	TESTING OF HYPOTHESIS-II	CLASSES: 08

Student t-distribution, its properties; Test of significance sample mean and population mean, difference between means of two small samples. Snedecor's F- distribution and its properties. Test of equality of two population variances. Chi-square distribution, its properties, Chi-square test of goodness of fit. One way ANOVA

UNIT - V**STOCHASTIC PROCESSES AND MARKOV CHAINS****CLASSES: 08**

Introduction to Stochastic processes- Markov process. Transition Probability, Transition Probability Matrix, First order and Higher order Markov process, n-step transition probabilities, Markov chain, Steady state condition, Markov analysis.

TEXT BOOKS

1. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.
2. Probability and Statistics for Engineers by Richard Arnold Johnson, Irwin Miller and John E. Freund, New Delhi, Prentice Hall.
3. Probability and Statistics for Engineers and Sciences by Jay L. Devore, Cengage Learning

REFERENCE BOOKS

1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. Fundamentals of Mathematical Statistics by S.C. Gupta & V.K. Kapoor, S. Chand
3. Introduction to Probability and Statistics for Engineers and Scientists by Sheldon M. Ross, Academic Press

WEB REFERENCES

1. https://www.efunda.com/math/math_home/math.cfm
2. <https://www.ocw.mit.edu/resources/#Mathematics>
3. <https://www.sosmath.com/>
4. <https://www.mathworld.wolfram.com/>

E-TEXT BOOKS

1. <https://www.e-booksdirectory.com/details.php?ebook=10166>

MOOCS COURSE

1. <https://swayam.gov.in/>
2. <https://onlinecourses.nptel.ac.in/>